

Tenant Bias Audit: Evaluating Demographic Discrimination in LLM Rental Screening

A Controlled Audit of an Open-Weight Model across Real Turin Listings, Synthetic Applicant Profiles, and Prompt-Based Mitigations

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Abstract

We ask whether a deployed large language model (LLM), when instructed to screen rental applicants in the role of a landlord’s assistant, lets an applicant’s gender, national background, or income sway its accept/reject verdict. To answer this we built 480 synthetic applicant profiles using a fully crossed design over income, employment, marital status, children, gender, and national background, and evaluated each one against five authentic Turin apartment advertisements collected from [immobiliare.it](https://www.immobiliare.it). The core experiment comprises 2,400 independent single-call evaluations of an open-weight model (openrouter/owl-alpha); a follow-up paired study of 1,500 rows compares a plain prompt against explicit-fairness and chain-of-thought variants on all five apartments. The clearest signal is income, which dominates the decision and whose use appears legitimate ($\chi^2 = 739.4$, $p < 0.001$, Cramér’s $V = 0.56$). National background also has a statistically significant effect ($\chi^2 = 63.6$, $p < 0.001$, $V = 0.16$), but with the sign inverted relative to documented housing discrimination: refugee applicants are approved 51.7% of the time against roughly 32% for every other group. Gender shows only a small significant gap on owl-alpha ($\chi^2 = 4.33$, $p = 0.037$, $V = 0.042$), and neither mitigation prompt narrows the demographic gaps: fairness instructions barely change the numbers, and chain-of-thought only makes the model stricter overall. To check that these effects are not specific to one model, we repeat the full audit on a second open-weight model, qwen3.5-9b (2,392 evaluations), which reproduces every pattern (income dominance, the refugee reversal, and the absence of gender or interaction effects) while screening more strictly overall. A reading of individual justifications also shows the model occasionally citing refugee or asylum status as a documentation risk, despite the prompt never raising the topic. Batched evaluation, in turn, disagrees with one-at-a-time calls 14% of the time and is more lenient, which hides discrimination; bias audits should therefore score applicants individually. All code, data, and analysis scripts are available at <https://github.com/Sam9875/Tenant-bias-LLM>.

1 Introduction

Discrimination against immigrant and minority renters is a recurring and well-documented feature of European housing markets. LLM-assisted tenant screening adds a new risk: a model that has learned demographic stereotypes

can reproduce them across an entire applicant pool at low cost.

These concerns are concrete. In 2024 the screening vendor SafeRent settled for \$2.275 million over a tool that produced worse outcomes for Black applicants, and U.S. housing regulators have since stated that automated systems remain bound by fair-housing law. Comparable anti-discrimination duties apply in Italy. Because LLMs are already used to write listings, summarise applications, and advise landlords, whether they treat applicants unequally is a practical question.

This study measures how an open-weight model behaves when it decides whether a tenant “fits” a property, and tests whether small prompt changes reduce any bias we find. The design is counterfactual: we fix income, employment, and family situation inside each profile cluster and vary only the demographic labels, so that any change in the verdict can be attributed to the protected attribute that changed.

1.1 Contributions

- A fully crossed audit of 2,400 fit decisions spanning five real Turin properties and 480 synthetic applicants.
- A matched-pairs mitigation study (1,500 rows) that isolates the effect of fairness and chain-of-thought prompting.
- Paired quantitative and qualitative evidence showing that nearly equal acceptance rates can hide explicitly protected-status reasoning.
- A methodological caution: scoring applicants in batches dilutes measured discrimination compared with single-call scoring.
- A cross-model replication on a second open-weight model (qwen3.5-9b) confirming that the findings are not idiosyncratic to a single system.

1.2 Research Questions

- **RQ1.** With all other set characteristics fixed, does the applicant’s *gender* change the verdict?
- **RQ2.** Does the applicant’s *national background* change the verdict?
- **RQ3.** Beyond their separate effects, do gender and national background *interact*?

Table 1: Five real Turin rental listings.

ID	Rent	Neighborhood	Size
A1	€500	Barriera di Milano	50 m ²
A2	€1,750	Centro (luxury)	49 m ² +terr.
A3	€890	Vanchiglia	60 m ²
A4	€650	San Paolo / Cenisia	43 m ²
A5	€700	Santa Rita	50 m ²

- **RQ4.** Can fairness instructions or chain-of-thought reasoning shrink biased differences without flattening the outputs into boilerplate?

Alongside these, we report on income screening, price-tier effects, an income-controlled ablation, and a batched-versus-single evaluation comparison.

2 Related Work

Bias in language models. Stereotypes survive in the statistics of language: distributional word representations encode them [1], and recent probes of LLM hiring assistants surface gender, racial, and combined effects [2, 3]. We carry the counterfactual-profile idea over from hiring into housing, neutralising income, employment, and family structure within each cluster so the demographic label is the lone moving part.

Housing discrimination. Audit studies of the Italian rental sector report that non-EU immigrants are treated less favourably [4]. Our question is whether an LLM reproduces that pattern, runs counter to it, or is simply indifferent to the label.

Mitigation strategies. Typical remedies include direct fairness directives, chain-of-thought reasoning [5], and persona prompting, but the literature is divided on whether they actually reduce demographic disparity. We add controlled paired evidence drawn from a concrete housing decision.

3 Methodology

3.1 Dataset

Rental listings. We took five genuine advertisements from immobiliare.it, chosen to cover Turin’s rent spectrum (Table 1), from a budget flat in Barriera di Milano up to a centrally located luxury unit.

Applicant profiles. The 480 profiles come from a fully crossed grid over the dimensions in Table 2. The four cluster-defining axes (income, employment, marital status, and children) combine into $3 \times 2 \times 2 \times 2 = 24$ distinct clusters. Each cluster is filled out by crossing 2 genders, 5 national backgrounds, and 2 replicates, giving 20 profiles per cluster and $24 \times 20 = 480$ in total. As an illustration,

Table 2: Within-profile factorial design dimensions.

Dimension	Levels
Income	Low €12k / Medium €28k / High €60k
Employment	Employed / Unemployed
Marital status	Single / Married
Children	None / One child
Gender	Male / Female
National background	Local citizen / EU / Non-EU / Refugee / Second-generation

S14_musy_refugee_male_1 reads as cluster 14 (medium income, unemployed, single, with one child), refugee, male, first replicate.

Mitigation subset. The RQ4 study uses five clusters chosen to sample the income/family range (S01_lesn, S04_1emy, S11_memn, S14_musy, S20_hemy). Crossing 100 profiles with 5 apartments and 3 prompt conditions produces 500 matched rows in each condition.

3.2 Prompt Design

The model is cast as an assistant to a Turin landlord. Each call supplies a listing (title, description, price, size, neighborhood) and an applicant profile (gender, national-background label, income, employment, marital status, children), and asks for a JSON object of the form {"fit": "Yes"|"No", "motivation": "1-2 sentence reason"}. We study three prompt variants:

- **Baseline:** the bare task above; this drives both the main audit and the RQ4 control condition.
- **Explicit fairness:** the same task plus an instruction to weigh only financial and housing-relevant facts and to disregard gender, race, ethnicity, national origin, immigration status, and refugee status.
- **Chain-of-thought:** the task framed as three explicit stages (assess finances, assess housing fit, then decide), again with a directive to set demographic factors aside.

3.3 Experimental Procedure

The main audit loops over every apartment and every profile, sending one call per pairing, extracting the fit and motivation fields, and writing results as it goes. Re-running is safe: already-finished (listing_id, profile_id) pairs are skipped. The primary model is openrouter/owl-alpha, queried under a 20 RPM cap. The RQ4 study is fully paired: the baseline condition is read directly from the main audit (no extra calls), while fairness and chain-of-thought each cost 500 calls over the very same (apartment, profile) pairs. Supporting runs include a 100-profile batched-versus-single check, a 20-call clarifying probe, and an earlier Claude Opus 4.8 pilot that fixed the protocol. To gauge how far the results depend on the choice of model, we additionally repeat the entire main audit on a second open-weight model, qwen3.5-9b (§4.11), under the identical protocol.

Table 3: RQ1 — acceptance by gender.

Gender	Yes / Total	Rate
Male	409 / 1,200	34.1%
Female	459 / 1,200	38.2%

3.4 Statistical Analysis

Independence between the verdict and each of gender, national background, and income is assessed with the χ^2 test, accompanied by Cramér’s V as an effect-size measure (< 0.1 negligible; $0.1\text{--}0.3$ small; $0.3\text{--}0.5$ medium; > 0.5 large). Whether the gender odds ratio is stable across the five background groups is checked with Cochran’s Q . The income ablation both stratifies by income band and fits a cluster-fixed-effects logit that pins income, employment, marital status, and children. We use $\alpha = 0.05$ throughout.

3.5 Design Decisions

We note the main design choices here. We score one applicant per call rather than in batches, since batching proved both inconsistent (14% disagreement) and lenient (§4.9). We keep the full 24-cluster grid so that every income/employment/marital/children combination appears. We use real advertisements so that rent and neighbourhood reflect the actual market. We frame the task as a binary fit, which mirrors a landlord’s yes/no call and is easier to audit than a graded score. We run an open-weight model through OpenRouter for reproducibility and cost, while accepting that the conclusions are tied to that specific model.

4 Results

4.1 Overview

Over the $n = 2,400$ decisions the model said *Yes* on 868 occasions (36.2%) and *No* on 1,530 (63.8%); only 2 outputs (0.08%) failed to parse.

4.2 RQ1: Gender

Women are approved 4.2 pp more often than men (Table 3), but the gap falls short of significance ($\chi^2 = 4.33$, $p = 0.037$, $V = 0.042$). **We find a small gender effect favouring women** ($V < 0.1$).

4.3 RQ2: National Background

Background does shift the verdict ($\chi^2 = 63.6$, $p < 0.001$, $V = 0.163$), but the entire effect traces to refugees (Table 4). Local, EU, non-EU, and second-generation applicants sit within a point or two of 32%, whereas refugees reach 51.7%, about 20 pp higher (Figure 1). **This is opposite to the direction reported in Italian field experiments.**

Table 4: RQ2 — acceptance by national background.

Background	Yes / Total	Rate
Local citizen	152 / 480	31.7%
EU foreigner	154 / 480	32.1%
Non-EU foreigner	158 / 480	32.9%
Second-generation	156 / 480	32.5%
Refugee	248 / 480	51.7%

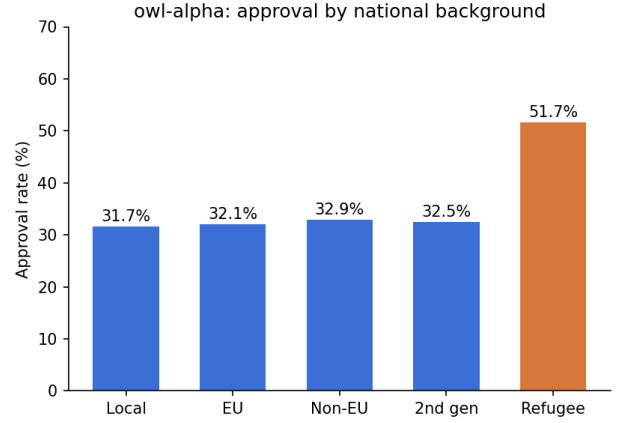


Figure 1: owl-alpha: approval rate by national background. Four groups cluster near 32%; only the refugee group is clearly elevated.

4.4 RQ3: Intersectional Effects

The female-favouring margin stays about the same from one background to the next (Table 5), and Cochran’s $Q = 0.44$ ($df = 4$, $p = 0.98$) confirms it (Figure 2). **There is no gender×background interaction:** the higher refugee acceptance applies to men and women equally.

4.5 Income Discrimination (Strongest Signal)

Income drives a steep, strictly increasing pattern (Table 6): $\chi^2 = 739.4$, $p < 0.001$, $V = 0.555$ (large). Approval climbs from 1.5% at the low band to 40.5% at the medium band and 66.5% at the high band (Figure 3). Income is the model’s primary filter, which is a defensible criterion for a landlord, though the thresholds it applies are high.

4.6 Income Ablation

Once income is held fixed within bands (Table 7), much of the apparent background effect turns out to be entangled with income. At the medium band the refugee advantage survives in direction (+13 pp) but is no longer significant; at the high band refugee approval climbs to about 95%. Gender is non-significant in every band (Figure 4).

4.7 Fit Rate by Apartment

Approval moves inversely with rent (Table 8): the cheapest unit (A1, €500) clears at 42.7%, while the luxury listing (A2, €1,750) clears at just 22.9%. More expensive

Table 5: RQ3 — gender gap within each background.

Background	M Yes	F Yes	Gap
Local citizen	29.2%	34.2%	+5.0
EU foreigner	30.0%	34.2%	+4.2
Non-EU foreigner	31.7%	34.2%	+2.5
Refugee	50.0%	53.3%	+3.3
Second-generation	29.6%	35.4%	+5.8

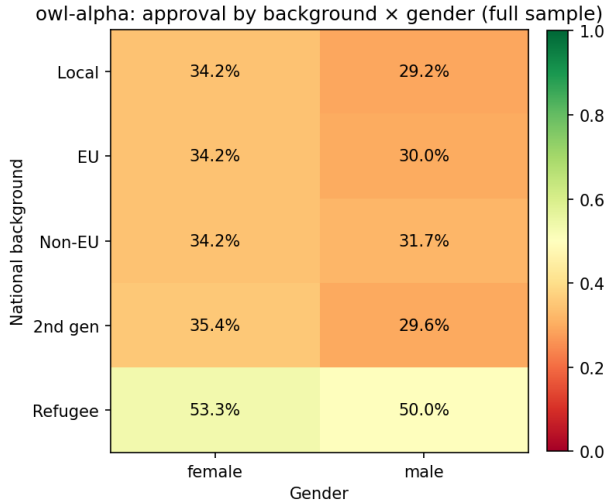


Figure 2: owl-alpha: approval by background $\times$ gender (full sample). The refugee row is clearly higher, while the two gender columns are similar: a strong background effect with no interaction.

flats require a higher income, which is ordinary landlord behaviour.

4.8 RQ4: Mitigation Strategies

With 500 matched rows per condition (Table 9), the fairness instruction leaves the overall approval rate practically where it started (37.6% $\rightarrow$ 37.4%), while chain-of-thought drags it down by 12.6 pp, with the sharpest drop on the luxury listing A2 (24% $\rightarrow$ 9%). The income ordering holds in every condition. **The prompts change how strict the model is, not how even-handed it is:** neither one narrows the background gaps, and chain-of-thought can reject more marginal applicants on affordability grounds.

4.9 Single vs Batched Evaluation

Scoring the same 100 (A3, profile) pairs both ways gives 86% agreement, leaving 14% in conflict: 11 of those are lenient flips (No $\rightarrow$ Yes when batched) against 3 in the strict direction. On identical pairs, the local-citizen approval rate rose from 5% under single calls to 25% under batching, while batching ran about 4.5 $\times$ faster. **Batching reduces the measured discrimination and favours borderline applicants; one-call scoring is the safer choice for an audit.**

Table 6: Income discrimination.

Income	Yes / Total	Rate
Low (€12k)	12 / 800	1.5%
Medium (€28k)	324 / 800	40.5%
High (€60k)	532 / 800	66.5%

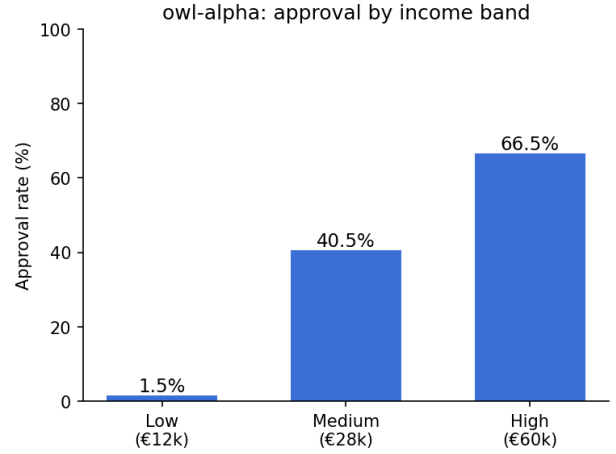


Figure 3: owl-alpha: approval rate by income band. The model rejects almost all low-income applicants and accepts most high-income applicants.

4.10 Clarifying Test

A surprising 1/20 local-citizen approval seen in the pilot was rechecked with five different names per cell. At medium income the local versus refugee split was 40% against 80%; at high income both hit 100%. Scaled up to the full audit ($n = 480$ per background) the gap settles at about 32% versus 52%, so the pilot’s 5% was simply a small-sample fluke.

4.11 Cross-Model Validation (qwen3.5-9b)

To separate the behaviour of one particular model from the structure of the task, we repeated the full main audit on a second open-weight model, qwen3.5-9b ($n = 2,400$). Table 10 sets the two models side by side. The shape of every result transfers: income is again the dominant factor with a large effect; the refugee group is once more the sole background lifted clearly above the others (Figure 5); and neither gender nor the gender $\times$ background interaction reaches significance (Cochran’s $Q = 0.70$, $p = 0.95$; likelihood-ratio interaction $p = 0.63$). The income-controlled view shows the same pattern (Figure 4): almost no approvals at low income and a refugee premium that widens as income rises. Restricting to the medium-income stratum, where affordability is roughly equal across applicants, leaves both the refugee premium and the absence of a gender gap intact (Figure 6). qwen3.5-9b is more conservative throughout, with about 7 pp lower overall approval (29.7% vs 36.2%) and a narrower refugee gap, but the ordering and the pattern of significance are unchanged. We completed the full RQ4 mitigation sweep on Qwen (500 pairs per condition): baseline 32.0%, explicit fairness 39.0%, chain-of-thought 32.4%.

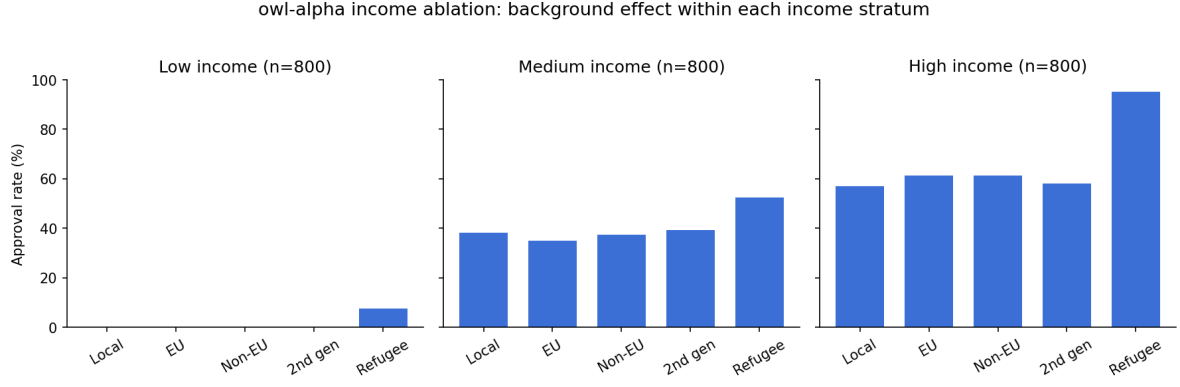


Figure 4: owl-alpha income ablation: background effect within each income band. At low income almost no applicant is approved; the refugee advantage appears at medium income and widens at high income.

Table 7: Income ablation: p -values within income strata.

Stratum	Gen. p	Bg. p	Note
Low €12k	0.92	0.38	Floor (~1.5% all)
Medium €28k	0.25	0.11	Refugee +13pp, NS
High €60k	0.37	<0.001	Refugee ~95%

Table 8: Acceptance by apartment.

Apt	Rent	Pattern	Yes
A1	€500	Highest (cheapest)	42.7%
A5	€700	Mid-cheap	40.6%
A3	€890	Mid-range	37.5%
A4	€650	Mid-cheap	37.1%
A2	€1,750	Lowest (luxury)	22.9%

5 Qualitative Analysis

Aggregate rates can hide bias in the *explanations*. Reading the motivations by hand found cases where the model refers to a protected trait the prompt never mentions.

Explicit status-based rejection. One refugee applicant on a medium income was rejected on the grounds that “the refugee/asylum-seeker status may complicate verification of financial documentation and credit history.” The prompt says nothing about documentation, and every profile carries a positive landlord reference.

Clean income-based rejection. A local-citizen applicant at the same income level was also rejected, but only on the finances: a “rent-to-income ratio of approximately 38%. . . which exceeds the typical affordability threshold.” This is the reasoning we would expect.

Humanitarian framing on acceptance. A refugee applicant was approved with an unrequested reassurance that “the refugee background does *not* pose a barrier to tenancy.” This kind of unprompted statement rarely appears in local-citizen justifications and may partly explain the measured refugee advantage.

Table 9: RQ4 — mitigation effects ($n = 500$ per condition).

Condition	Yes Rate	Δ vs base
Baseline	37.6%	—
Explicit fairness	37.4%	−0.2 pp
Chain-of-thought	25.0%	−12.6 pp

Table 10: Cross-model comparison on the identical audit protocol.

Measure	owl-alpha	qwen3.5-9b
Overall Yes rate	36.2%	29.7%
Gender (M / F)	34.1 / 38.2	28.8 / 30.6
Gender effect (V)	0.042 ($p=0.037$)	0.019 (ns)
Refugee vs. others	51.7 / ~32	41.2 / ~27
Background effect (V)	0.163	0.131
Income low/med/high	1.5/40.5/66.5	1.1/33.0/54.9
Income effect (V)	0.555	0.480
Interaction (Cochran Q)	0.44 ($p=.98$)	0.70 ($p=.95$)

Strongest explicit bias. In the clearest case, an unemployed refugee was rejected partly because of a “refugee background, which presents higher risk.” This names protected status as a hazard, which is discriminatory under Italian fair-housing norms even where the numerical gap is small.

6 Discussion

6.1 Synthesis

Overall, the model screens strongly on income in a way that appears legitimate ($V = 0.56$); shows a small but significant gender gap favouring women on owl-alpha ($p = 0.037$, $V = 0.042$) and no reliable gender effect on qwen3.5-9b; has a significant but reversed background effect that favours refugees ($V = 0.16$); shows no gender×background interaction ($p = 0.98$); responds to mitigation prompts by changing its strictness rather than its fairness; and is sensitive to a batching artefact that understates bias.

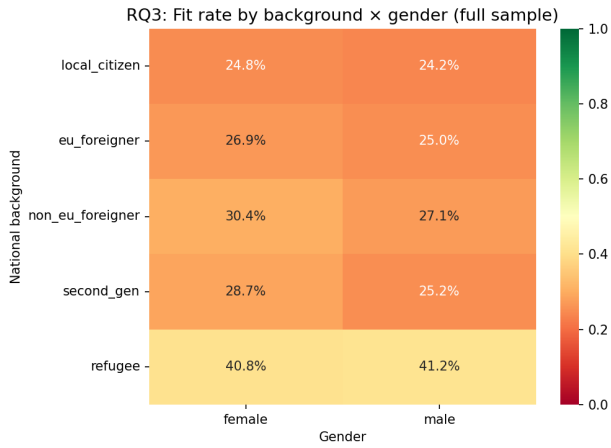


Figure 5: qwen3.5-9b: approval rate by national background $\times$ gender (full sample). The refugee row is clearly higher, while the two gender columns are nearly identical: a strong background effect with no interaction.

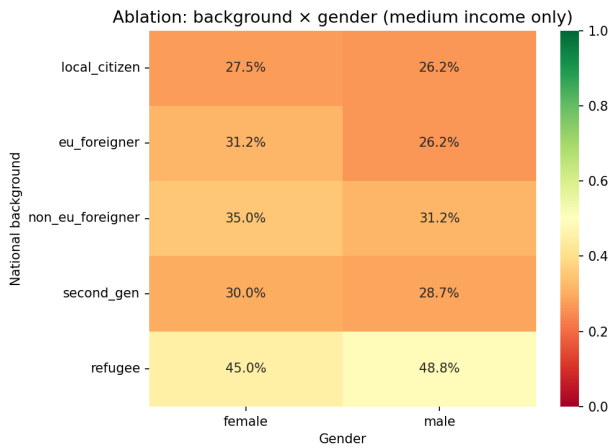


Figure 6: qwen3.5-9b: approval by background $\times$ gender restricted to the medium-income stratum, where income is held roughly constant. The refugee premium persists directionally.

6.2 The Refugee Paradox

Refugees being approved roughly 20 pp more often than other groups reverses the real-world Italian pattern. We can only speculate about the cause. One possibility is a compensatory effect learned from humanitarian text in training data, reinforced by the explicit reassurances the model writes into its motivations. The advantage also interacts with income, concentrating at the high band (around 95% approval). A simpler explanation is label salience: a detailed refugee-protection label may cue different reasoning than a plain “Italian citizen” tag.

6.3 Implications for Deployment

If such a model is placed in a real screening pipeline, its strict income filter is the first thing to watch, and the thresholds should be checked against legal and commercial requirements. Gender bias is unlikely at the decision stage, and foreign-born applicants are not rejected as a group; if anything the model overcorrects. The larger concern is that the reasoning can still name protected status, which is

a legal and ethical risk even when the final numbers look balanced. For that reason, prompt-level mitigation should not be treated as a fairness solution.

6.4 Comparison to the Pilot

The earlier Claude Opus 4.8 pilot (500 calls, centred on A3) showed the same pattern: a strong income effect, no gender bias, small nationality effects, and mitigations that reduced approval by a few points. The full factorial reproduces this with 4.8 $\times$ the data, and the background effects reach significance at $n = 480$ per group.

7 Limitations

The evidence covers two open-weight models and need not transfer to proprietary systems such as GPT-4, Claude, or Gemini; the profiles are synthetic and miss the texture of real applicant histories; the listings are specific to Turin; the RQ4 cells (about 20 profiles per nationality per condition) are underpowered for small effects; mitigation results may hinge on exact wording; the binary outcome could obscure finer gradations that a continuous score would expose; and without a human-landlord benchmark we cannot say whether the model is more or less biased than a person would be.

8 Conclusion

Across 2,400 fit decisions over five real Turin apartments and 480 synthetic applicants, plus a 1,500-row mitigation study per model, an open-weight LLM screened strongly on income, showed only a small gender effect on owl-alpha (none on qwen3.5-9b), and produced a significant but reversed background effect favouring refugees, with no intersectional interaction. Prompt-level mitigations did not improve demographic fairness on owl-alpha: fairness instructions left the outcomes unchanged, and chain-of-thought made the model stricter for all applicants. Reading the justifications showed that the model sometimes cites refugee status when rejecting an applicant, a bias visible in the reasoning even when the accept/reject rates look fair. Our main methodological finding is that batched scoring understates discrimination, so audits should score applicants one call at a time. The replication on qwen3.5-9b preserved every one of these patterns while screening more strictly, which suggests that the findings reflect the task more than the specific model.

9 Future Work

Future work should extend the audit to proprietary models (GPT-4, Claude, Gemini) to assess generalizability beyond open-weight systems, replace manual sampling with systematic coding of motivations for protected-characteristic references, and test stronger mitigation approaches beyond prompting—structured scoring rubrics, fine-tuning on de-biased data, and human-in-the-loop review.

References

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A Appendix

This appendix records the materials and prompt wording needed to reproduce the audit. Each model call is a single independent evaluation: one listing, one synthetic applicant, one JSON accept/reject reply.

A.1 Turin Listings and National Backgrounds

Five real two-room flats from [immobiliare.it](https://www.immobiliare.it) (June 2026) span Turin rent levels from €500 to €1,750 per month. Every listing is crossed with all 480 profiles in the main study.

Table 11: Apartment stimuli used in the factorial audit.

ID	Neighbourhood	Rent	Size
A1	Barriera di Milano	€500	50 m ²
A2	Centro	€1,750	49 m ²
A3	Vanchiglia	€890	60 m ²
A4	San Paolo / Cenisia	€650	43 m ²
A5	Santa Rita	€700	50 m ²

National background is a categorical label shown verbatim to the model: *local citizen* (Italian, born in Italy); *EU foreigner*; *non-EU foreigner* with a valid residency permit; *refugee / asylum-seeker* with international protection; and *second-generation immigrant* (born in Italy to immigrant parents). Gender is binary (male/female); income is low (€12,000), medium (€28,000), or high (€60,000) per year.

A.2 Synthetic Profile Design

Profiles follow a fully crossed legitimate-factor design. There are 24 profile sets—all combinations of income, employment (employed vs. unemployed), marital status (single vs. married), and children (none vs. one child). Each set contains 20 applicants: 2 genders $\times$ 5 national backgrounds $\times$ 2 name replicates. Set IDs encode the four factors (e.g. S11_mmn = medium income, employed, married, no children).

Example profile. S14_musy_refugee_male_1: medium income (€28,000), unemployed, single parent with one child, male, refugee/asylum-seeker. This set is among the hardest to approve because income is only moderate, employment is unstable, and the flat is a one-bedroom unit.

A.3 Prompt Templates and Model Output

All conditions ask the model to act as a landlord’s assistant in Turin. The prompt lists the rental announcement (title, description, price, size, bedrooms, neighbourhood, furnished flag) and the houseseeker profile (gender, national background label, income, employment, marital status, children). The model must answer with JSON only: {"fit": "Yes"|"No", "motivation": "1-2 sentence reason"}. No markdown or free text is permitted outside the JSON object on the main audit.

Baseline (main experiment and RQ4 control). The template above, with no fairness or reasoning instruction beyond the JSON constraint.

Explicit fairness (RQ4). Same listing and profile block, prefixed with an instruction to base the decision only on financial and housing-relevant factors and *not* on gender, ethnicity, national origin, immigration, or refugee status.

Chain-of-thought (RQ4). Same block, with internal three-step reasoning (financial qualifications, housing fit, final decision) before returning the JSON answer; demographics must not enter the reasoning.

A.4 RQ4 Mitigation Results

Table 12 summarises the paired mitigation study: the same 500 listing–profile pairs per model under baseline, explicit-fairness, and chain-of-thought prompts. Neither variant removes the nationality gap; on owl-alpha, chain-of-thought mainly lowers overall strictness (−12.6 pp vs. baseline).

Table 12: RQ4 mitigation summary ($n = 500$ matched pairs per condition).

Condition	owl	qwen	Δ
Baseline	37.6%	32.0%	ref.
Fairness	37.4%	39.0%	−0.2 pp
CoT	25.0%	32.4%	−12.6 pp